

Marine Cases CV

13 mars 2026

Education

2025–2026	Master 2 in Mathematics, Jussieu University Courses taken : Homology, cohomology and sheaves, Varieties, Schemes 1 and 2, Complex geometry and Hodge theory, Homotopy 1 and 2, and perverse sheaves 1 and 2, character sheaves.
2024–2025	Master 1 in Mathematics, Jussieu University Courses taken : 1st semester : Differential Geometry, Commutative Algebra, Complex Analysis 2nd semester : Galois Theory, Lie Theory, Algebraic Number Theory, Algebraic Topology, Riemann Surfaces
2023–2024	Bachelor 3 in Intensive Mathematics (advanced algebra track), Jussieu University
2021–2023	Bachelor in Mathematics, Paris–Saclay University
2020–2021	MPSI at Janson (computer science option) Reorientation during the year during COVID
2017–2020	French & Spanish double baccalaureate (OIB), Lycée International of Saint-Germain-en-Laye Highest honors in both.

Projects in M2

Master thesis : research project	Computing the categorical trace of derived geometric satake on $\mathbb{F}_q$, under the supervision of Prof. Guglielmo Nocera and Prof. Simon Riche
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Projects in M1

First year of master's thesis on Deligne-Lustig theory	Learning Deligne-Lustig theory and character sheaves to have an example of what is categorification. Supervised by Prof. Eric Vasserot and Prof. Dennis Gaitsgory.
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Projects in L3

A bit of Logic	Project on the Löwenheim–Skolem theorems and introduction to model theory, with Prof. A. Deloro.
L3 Project Course	Paper on the Kubota–Leopoldt L -function, Jussieu–University of Copenhagen partnership, supervised by Prof. A. Betina.

Projects in L2

A bit of Commutative Algebra	On the Nullstellensatz and some basics of algebraic geometry, supervised by Prof. M. Cavicchi.
A bit of Homological Algebra	On acyclic models, the Salamander Lemma and triangulated categories, supervised by Prof. M. Livernet.
Dollar Game	Personal research with feedback from Prof. M. Baker. Preprint : https://arxiv.org/abs/2308.12787 .

Personal Information

Interests	Math, Figure skating, Reading, Piano, Climbing, Languages, Fencing, Horse riding (certified Galop 4)
Volunteering	Scouts (outreach activities and Christmas toy collection), French Red Cross
French	C2* ●●●●●● <i>Native</i>
Spanish	C2* ●●●●●● <i>Native</i>
English	C1* ●●●●○
German	B2* ●●○○○

Presentations

M1 TER — Seminar, october 2024	Talk on Lang’s Theorem
M2 Seminar, — december 2024	Talk on Pontryagin duality (Prof. J. Fresan)
Online — seminar on derived algebraic geome- try,27/11/2025	Introduction to IndCoh, following Gaitsgory-Rozemblyum

— Reading group on geometric class field theory, 7/12/2026	Talk on the Harder-Narasimhan filtration (wanted to do for G reductive but only had time to do Gln) following V.Pilloni's notes (planned the talk following chapter 5.2 of Gaitsgory-Lurie's Weil conjectures for function fields 1 too, to introduce the HN-Shatz filtration)
— M2 groupe de travail sur la correspondance de Riemman Hilbert, 19/01/2026	Proof of a first case of (complex) RH correspondance : equivalence of categories between complex local systems and complex vector bundles with a flat connexion

Schools + conferences

September 3–11, 2024	Topos in Mondovì, Mondovì, Italy
July 7–11, 2025	Theta Summer School, Norwich, UK
August 11–15, 2025	Conference in Memory of Yuri Manin, Bonn, Germany
August 18–22, 2025	Masterclass “Proof of the geometric Langlands conjecture”, University of Copenhagen, Denmark
August 25–29, 2025	Geometric Aspects of the Langlands Correspondence, Essen, Germany
September 8–12, 2025	Workshop on the Endoscopic Classification, in honor of James Arthur and Gérard Laumon, University of Chicago's Paris Center
September 17–19, 2025	Conference for Algebraic Geometry via Eurostar (CAGE), Leiden, Netherlands
September 24–26, 2025	Affine W-Algebras and Quantum Groups, Université Paris Cité, France
October 6–10, 2025	ReMoLD School : representations, motives and Langlands duality, Essen, Germany
December 8–12, 2025	Workshop “Geometry, Higher Structures, and Physics”, ICMS (Bayes Centre), Edinburgh, United Kingdom
December 16–18, 2025	Masterclass on n-shifted symplectic structures, Angers, France